

K-Means Clustering — Lab Assignment

Goal. Implement K-Means clustering from scratch and understand how the algorithm partitions unlabeled data into K clusters by minimizing intra-cluster variance. You will compute the objective, visualize convergence and cluster assignments, and automatically identify the optimal number of clusters via validation-based criteria. (No scikit-learn for the K-Means training itself.)

Formulas

K-Means Objective

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K) = \frac{1}{m} \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

where $c^{(i)}$ is the index of the centroid assigned to $x^{(i)}$.

Centroid Update

$$\mu_k = \frac{\sum_{i=1}^m \mathbf{1}\{c^{(i)} = k\} x^{(i)}}{\sum_{i=1}^m \mathbf{1}\{c^{(i)} = k\}}$$

Within-Cluster Sum of Squares (WCSS)

$$\text{WCSS}(K) = \sum_{k=1}^K \sum_{x^{(i)} \in C_k} \|x^{(i)} - \mu_k\|^2$$

Silhouette Coefficient

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \quad a(i) = \text{mean distance to points in same cluster}, \quad b(i) = \min_{\ell \neq c^{(i)}} \text{mean distance to cluster } \ell$$

Exercise 1 — K-Means (from scratch)

Task. Implement a K-Means learner on a dataset $X \in \mathbb{R}^{m \times n}$:

- (a) *Initialize centroids:* randomly select K examples as initial $\mu_1, \dots, \mu_K$.
- (b) *Assignment step:* for each $x^{(i)}$, set $c^{(i)} = \arg \min_k \|x^{(i)} - \mu_k\|^2$.
- (c) *Update step:* recompute each μ_k as the mean of points assigned to it.
- (d) *Repeat:* alternate (b)–(c) for a fixed number of iterations or until assignments stop changing.
- (e) *Convergence plot:* plot J versus iteration number.
- (f) *Visualization:* scatter plot of data (colored by cluster), mark centroids (e.g., “x”).

Numeric Evaluation (for Moodle/VPL)

Q1.1 — Assignment step

Dataset A: $X = \{(1, 1), (2, 1), (4, 3), (5, 4)\}$, initial centroids $\mu_1 = (1, 1)$, $\mu_2 = (5, 4)$. Assign each point to the nearest centroid (1-based labels).

Expected output

1 1 2 2

Q1.2 — Centroid update

Use assignments from Q1.1. Compute updated centroids (rounded to 2 decimals).

Expected output

1 = (1.50, 1.00)

2 = (4.50, 3.50)

Q1.3 — Cost after first iteration

With updated centroids from Q1.2, compute

$$J = \frac{1}{4} \sum_{i=1}^4 \|x^{(i)} - \mu_{c(i)}\|^2.$$

Expected output

J = 0.38

Q1.4 — Two iterations (fixed) & final values

Dataset B : $X = \{(0, 0), (0, 3), (10, 10), (11, 11)\}$, initial centroids $\mu_1 = (0, 0)$, $\mu_2 = (11, 11)$. Run exactly two iterations (assignment + update).

Expected output

1 = (0.00, 1.50) 2 = (10.50, 10.50)

sizes = [2, 2]

J = 1.38

Q1.5 — Convergence plot values

For Dataset B above, record J after each iteration.

Expected output

J_history = [1.38, 1.38]

Implementation Hints

- Use Euclidean distances; vectorize computations when possible.
- Randomly initialize from data points; fix a random seed for reproducibility. Re-seed empty clusters to a random point.
- Early stop if assignments do not change between iterations.
- For 2D plots, use the first two features; label axes; mark centroids with “x”.
- Round printed numeric outputs to 2 decimals for grading.

Sample Test Case

input =

6

1.0 1.5

1.5 2.0

5.0 8.0

6.0 9.0

1.0 0.6

9.0 11.0

K = 2

Iterations = 10

```
output =  
  
Final Centroids:  
1 = (1.17, 1.36)  
2 = (6.67, 9.33)  
Cluster Assignments: 1 1 2 2 1 2  
Final Cost J = 5.29
```